CLINICAL WORKFLOW & ARCHITECTURE SPECIFICATION

VisionXAI

Explainable AI for Early Diabetic Retinopathy Screening

A 3-Tier Multi-Modal Tele-Ophthalmology System for Rural Outreach

Rural Field Screening | Dual-Stage XAI | Fundus Imaging | Doctor Governance | Supabase Auth

Executive Summary & System Purpose

Key System Highlights & Capabilities

- 3-Tier Clinical Workflow: Strict role-based isolation between ASHA Field, Nurse Clinic, and Specialist Doctor.
- Dual-Stage Explainable AI: Tabular SHAP factor attributions at Tier 1 + Visual Grad-CAM heatmaps at Tier 2.
- Multi-Language Support: Real-time reactive language toggle supporting English, Marathi (·····), and Hindi (·····).
- Adaptive Healthcare UI: Professional Aptos Light typography, light purple white background, and dark mode.
- Omnichannel Patient Dispatch: Click-to-chat WhatsApp (wa.me) and SMS notification with personalized recommendations.
- Zero Cold-Start Friction: Client-side persistence via LocalStorage + optional Supabase authentication.

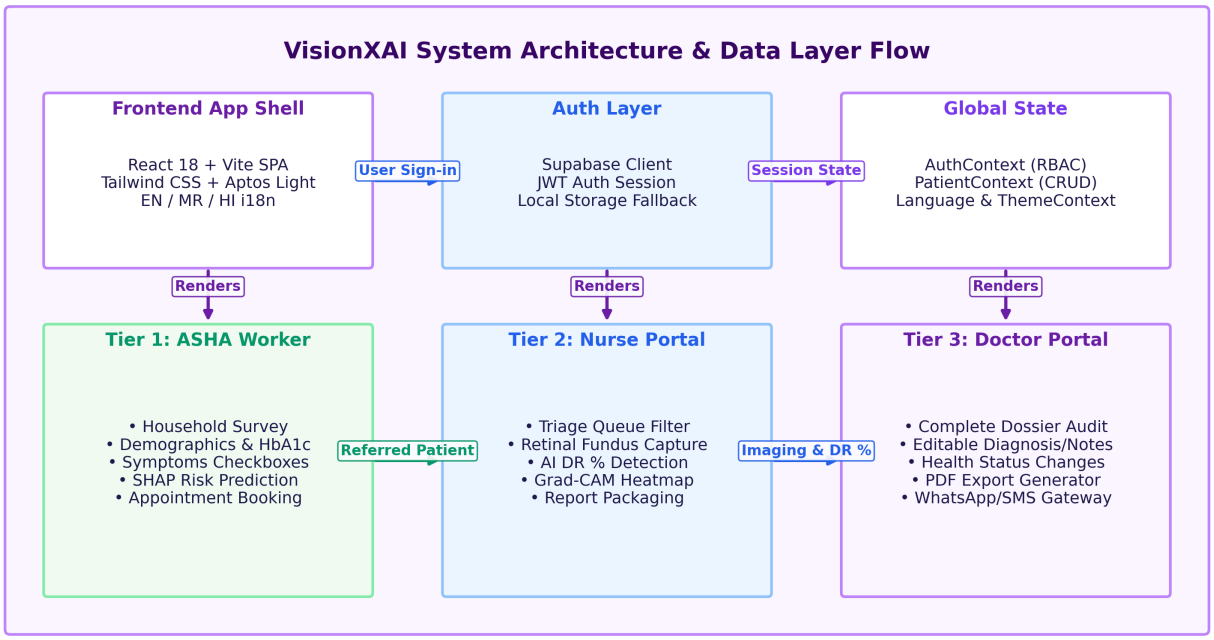
Document Version: 1.0.0 (Production Release)

Target Users: ASHA Healthcare Facilitators, Hospital Nurses, Consulting Ophthalmologists

Security Standard: HIPAA/DISHA-aligned data segregation with human-in-the-loop validation

System Architecture & Component Interactions

Full-stack client, state, and authentication layer design



Architecture Layer Breakdown

Role-Based Access Control (RBAC) & Data Compartmentalization

A fundamental safety requirement of the VisionXAI architecture is strict role segregation:

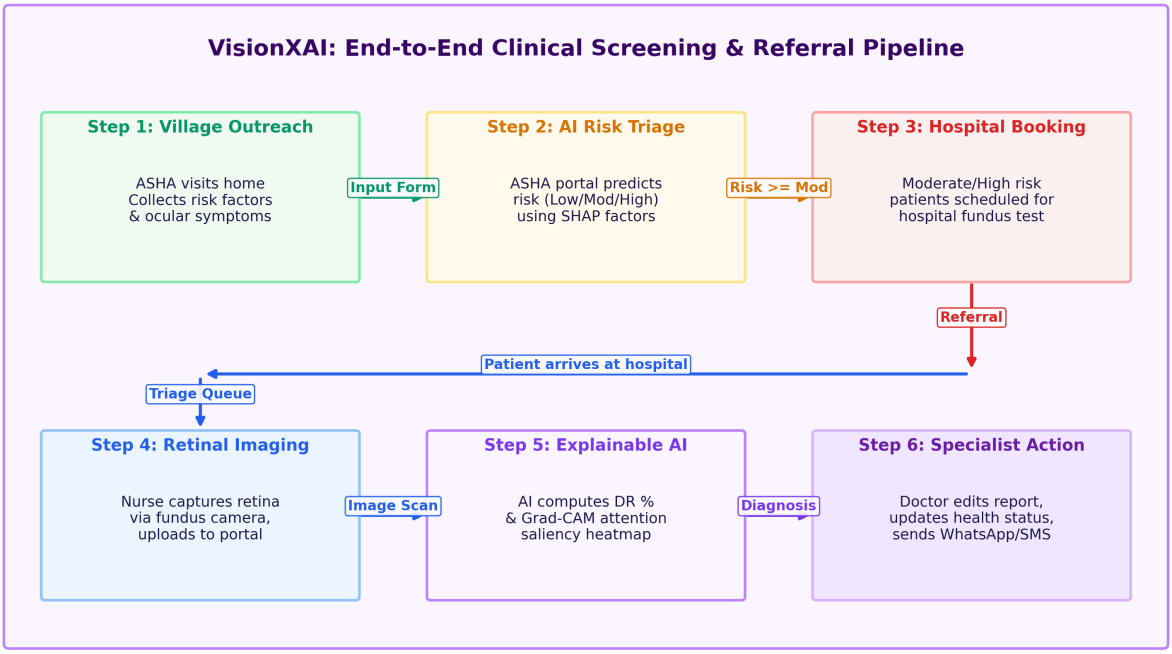
? No Direct Cross-Portal Switching: Users cannot jump directly between ASHA, Nurse, and Doctor portals from within a session. This prevents accidental cross-role data entry and maintains clean clinical boundaries.

? Clinical State Lifecycle: Patients transition through explicit states:
'Registered' (ASHA) ? 'Scheduled' (Appointment) ? 'Imaged' (Nurse) ? 'Diagnosed' (Doctor) ? 'Notified'.

? Privacy & Safety Controls: Every patient record includes deletion confirmation, eye image removal options, and audit trails to ensure compliance with clinical governance standards.

End-to-End Clinical Screening & Referral Pipeline

Step-by-step patient lifecycle from rural field visit to treatment

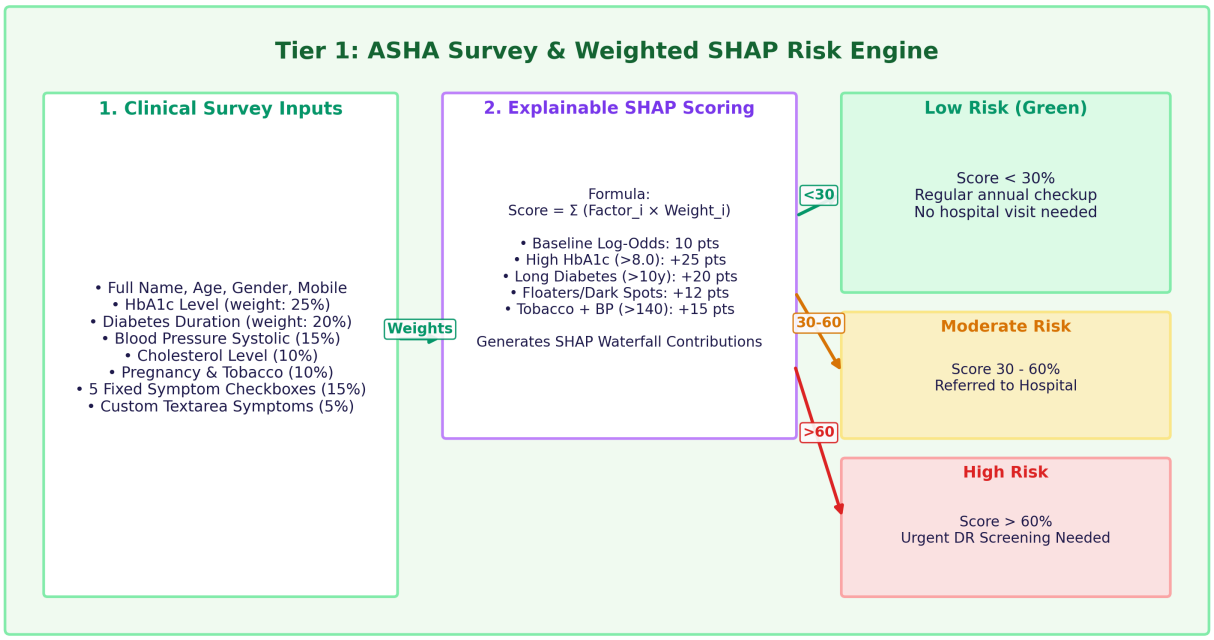


Phase-by-Phase Operational Protocol

Phase 1: Household Survey	ASHA Worker conducts rural outreach visits. Gathers patient demographics (Name, Age, Gender, Mobile), clinical history (HbA1c, BP, diabetes duration, cholesterol, pregnancy, tobacco use), and ocular symptoms.
Phase 2: AI Risk Assessment	ASHA clicks 'Predict Risk'. System evaluates factors using a weighted SHAP algorithm. Returns Low (<30%), Moderate (30-60%), or High (>60%) risk, complete with plain-language factor explanations.
Phase 3: Referral Scheduling	If risk is Moderate or High, AI flags mandatory DR hospital screening. ASHA schedules an appointment. Record is stored and automatically propagated to the hospital triage database.
Phase 4: Hospital Arrival & Imaging	Patient arrives at hospital clinic. Nurse accesses triage queue, verifies patient identity, and captures high-resolution retinal fundus photograph via specialized camera.
Phase 5: AI DR % & Grad-CAM Analysis	Nurse executes AI analysis. Deep learning model calculates Diabetic Retinopathy severity (0-100%) and renders a Grad-CAM heatmap highlighting suspicious lesions. Nurse generates clinical report.
Phase 6: Specialist Review & Editing	Ophthalmologist opens Doctor Portal, reviews full longitudinal dossier, edits clinical diagnosis and prescriptions, updates patient health status, and signs off.
Phase 7: Omnichannel Notification	Doctor dispatches report directly to patient's mobile device via WhatsApp (wa.me) or SMS, advising immediate follow-up actions and clinical instructions.

Tier 1: ASHA Field Survey & Explainable Risk Scoring

Community-based early risk prediction with weighted SHAP factor analysis



Input Parameters & Clinical Risk Weights

Clinical Factor	Weight	Clinical Significance & Rationale
Glycated Hemoglobin (HbA1c %)	25%	Primary biomarker; values above 7.0% increase DR microvascular damage exponentially.
Diabetes Duration (Years)	20%	Long-standing hyperglycemia (>10 years) strongly correlates with vessel leakage.
Blood Pressure (Systolic mmHg)	15%	Hypertension exacerbates retinal capillary pressure and microaneurysm rupture.
Serum Cholesterol (mg/dL)	10%	Elevated serum lipids correlate with hard exudate deposition in macular regions.
Pregnancy & Tobacco Use	10%	Pregnancy accelerates DR progression; nicotine induces retinal hypoxia.
Fixed Symptoms (Checkboxes)	15%	Blurry vision, floaters/spots, blank areas, faded colors, poor night vision.
Custom Manual Symptoms	5%	Free-text input for patient-reported itching, redness, or localized eye pain.

SHAP Explainability & Triage Recommendation Gate

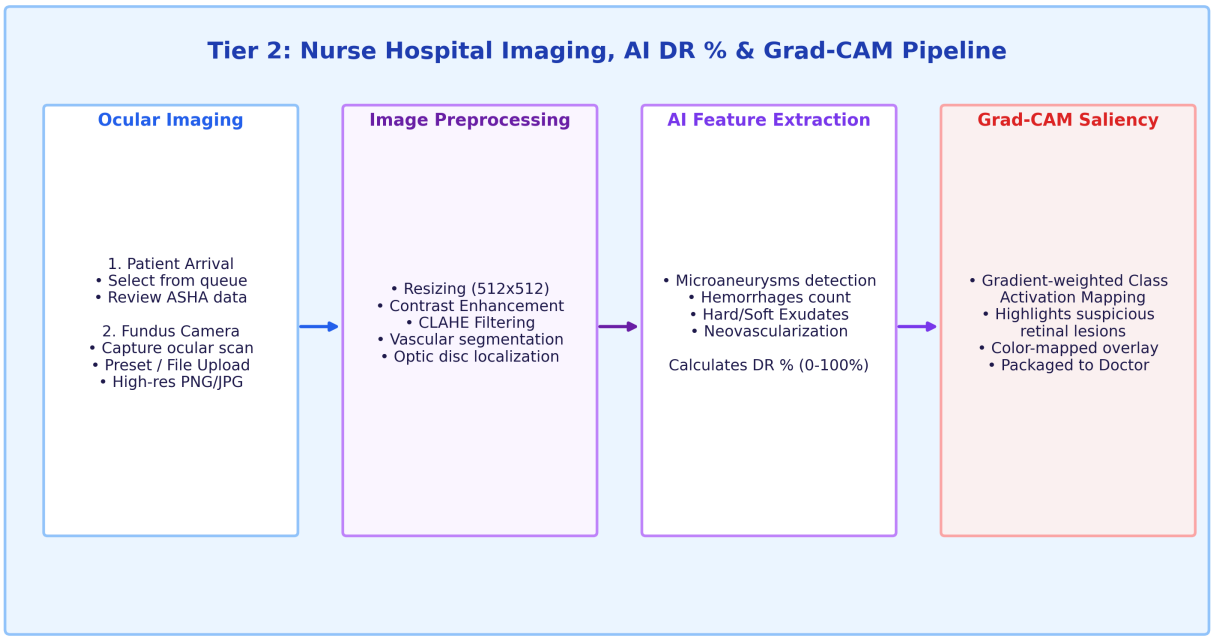
? Transparent Mathematical Formulation: The AI engine calculates composite risk using Shapley values:
Risk Score = Baseline + ? (Factor_i × Weight_i × Scaling_i)
Each factor produces a visible SHAP contribution bar, showing whether it increased or decreased risk.

? Triage Decision Logic:
- Low Risk (<30%): ASHA advises ongoing lifestyle management and regular annual review. No hospital trip needed.
- Moderate Risk (30-60%): AI flags early microvascular vulnerability. Recommends hospital fundus screening within 30 days.
- High Risk (>60%): Urgent notification. Immediate hospital appointment booked for comprehensive fundus photography.

? ASHA Dashboard Capabilities: Includes patient search, age/risk filtering, deletion options, and aggregate screening analytics.

Tier 2: Nurse Hospital Imaging & AI Detection Pipeline

High-resolution fundus image analysis, DR percentage calculation and Grad-CAM saliency



Retinal Imaging & Deep Learning Pipeline

1. Triage Queue & Patient Selection: When a referred patient arrives at the ophthalmology clinic, the nurse selects their profile from the triage queue. All prior survey data, risk scores, and symptoms collected by the ASHA worker are immediately visible.

2. Fundus Image Capture / Upload: The nurse captures high-resolution retinal images using standard fundus cameras. The portal supports both real-time file upload and clinical presets for offline simulation.

3. Computer Vision Preprocessing: Captured scans undergo illumination equalization, contrast-limited adaptive histogram equalization (CLAHE), and blood vessel enhancement to optimize lesion detection.

4. AI DR % Severity Engine: The vision model identifies hallmark diabetic microvascular lesions, including microaneurysms, blot hemorrhages, hard exudates (lipid deposits), and soft exudates (cotton-wool spots). It calculates an objective Diabetic Retinopathy percentage score (0-100%).

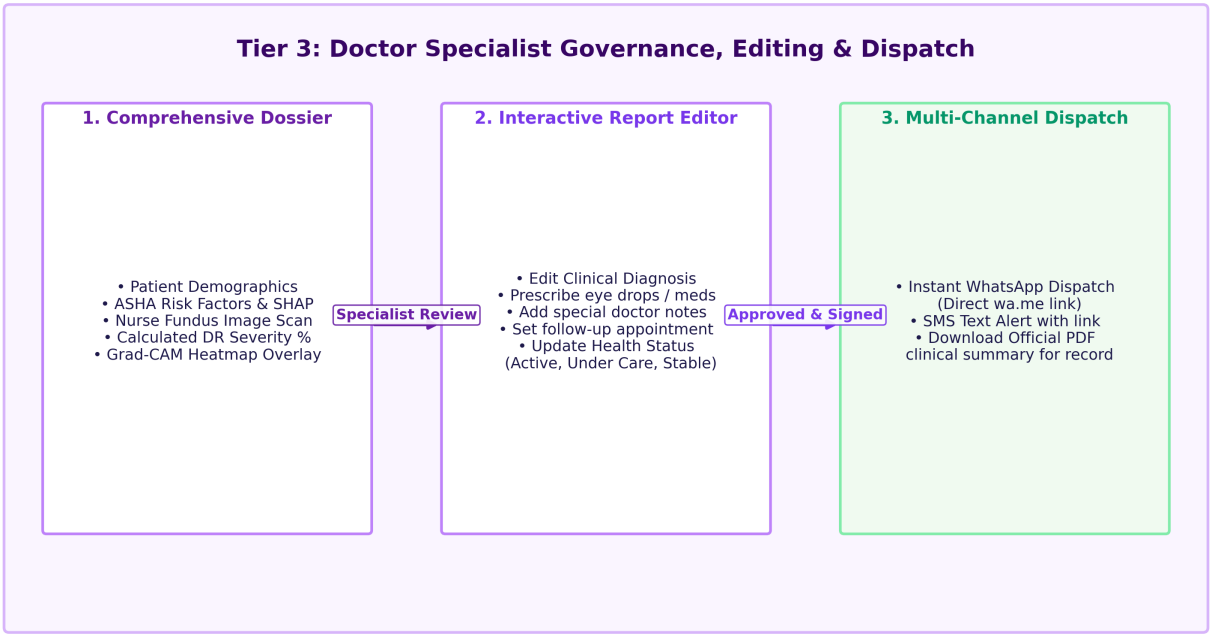
Grad-CAM Saliency Maps & Clinical Report Generation

? Grad-CAM Heatmap Visualization: To prevent blind trust in a black-box percentage, the AI engine produces a Gradient-weighted Class Activation Map (Grad-CAM). Using HTML5 Canvas rendering, the heatmap is overlaid directly onto the retinal image:
- Red & Orange Zones: High model activation ? areas with severe microaneurysms, hemorrhages, or exudates.
- Yellow & Green Zones: Moderate attention ? suspect vascular caliber changes or mild swelling.
- Blue & Transparent Zones: Normal retinal tissue, optic disc, or macula without lesions.

? Clinical Report Compilation: Nurse clicks 'Generate Report' to bundle patient demographics, ASHA survey factors, the retinal scan, calculated DR %, and Grad-CAM visualization into a clinical package routed to Tier 3 (Doctor).

Tier 3: Doctor Specialist Governance & Patient Communication

Ophthalmologist verification, interactive report editing, and omnichannel dispatch



Doctor Portal Capabilities & Clinical Workflow

1. Comprehensive Longitudinal Dossier: The Doctor Portal displays the unified patient record aggregating all data streams across Tier 1 (ASHA survey, SHAP risk), Tier 2 (fundus image, Grad-CAM heatmap, DR %), and past history.

2. Interactive Report Editor: The ophthalmologist maintains full clinical control and can edit:
? Clinical Diagnosis: Standard ICD-10 DR staging (Mild / Moderate / Severe NPDR or PDR, Macular Edema).
? Clinical Notes: Specialized remarks on laser photocoagulation, anti-VEGF therapy, or glycemic control.
? Prescriptions: Lubricants, anti-hypertensives, or retinal supplements.
? Follow-Up Interval: Scheduling next review (e.g., 3 months, 6 months, 1 year).

3. Patient Health Status Transition: Doctor updates official health status:
'Active' ? 'Under Treatment' ? 'Stable' ? 'Recovered' ? 'Critical'.

Omnichannel Patient Notification & Printable Clinical Report

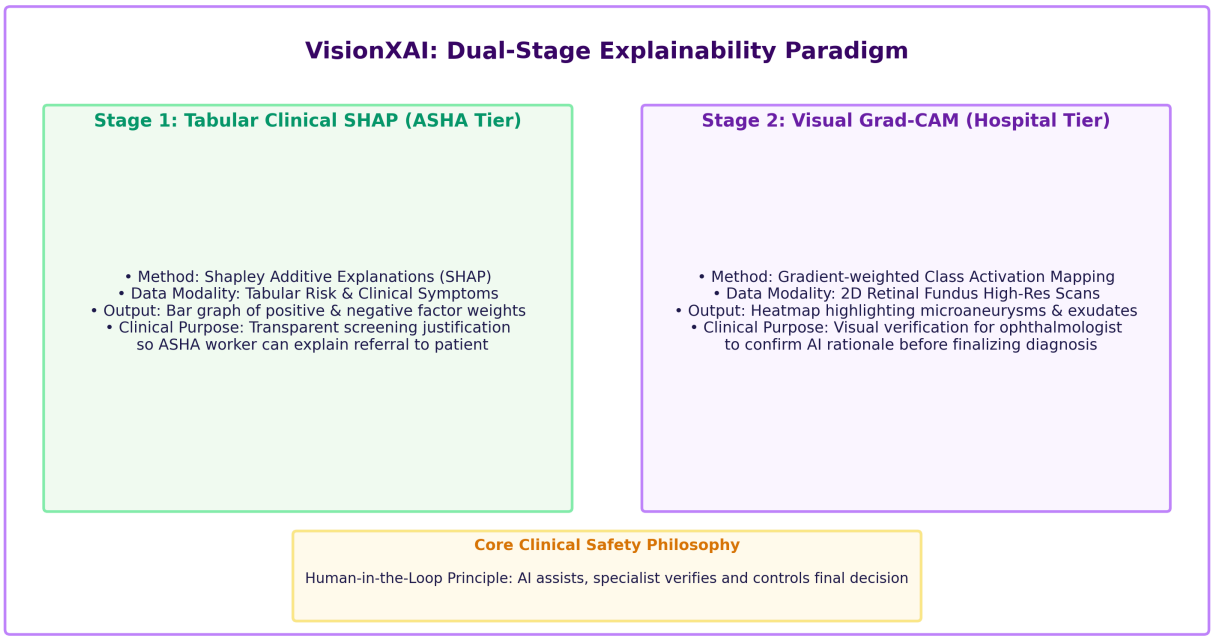
? Direct WhatsApp Dispatch (wa.me Integration): With one click, the doctor initiates a secure WhatsApp message pre-filled with the patient's mobile number, official diagnosis, severity grade, prescribed care, and hospital contact.

? Instant SMS Text Dispatch: For patients in rural villages without smartphones, an SMS alert provides the essential screening result, risk level, and next appointment date.

? PDF Clinical Report Export: Generates a printable, standardized diagnostic report with institutional branding, patient identifiers, retinal fundus images, Grad-CAM overlays, and doctor's signature line for physical hospital records.

Explainable AI (XAI) Framework & Clinical Safety

Dual-stage explainability paradigm ensuring medical transparency and clinical trust



Why Explainable AI is Critical in Ophthalmology

In clinical decision-making, conventional deep learning models act as 'black boxes' ? providing an output without revealing the rationale. In Diabetic Retinopathy screening, opaque predictions create severe risks:

? Erroneous Classifications: Artifacts such as lens dust, uneven illumination, or eyelashes can trick a CNN into falsely detecting hemorrhages or exudates, leading to unwarranted patient distress and hospital congestion.

? Lack of Clinician Trust: Ophthalmologists cannot ethically sign off on diagnoses without knowing which retinal regions the AI analyzed.

? Patient Communication: Rural patients are far more likely to comply with referrals when the ASHA worker can point to specific factors (e.g., 'your high HbA1c of 9.2% and floaters accounted for 45% of your risk').

Comparison: Tabular SHAP vs Visual Grad-CAM

Dimension	Tier 1: SHAP Engine	Tier 2: Grad-CAM Engine
Target User	ASHA Rural Health Worker	Hospital Nurse & Ophthalmologist
Data Modality	Tabular Risk & Symptoms	2D Retinal Fundus Scans
Mathematical Basis	Shapley game-theoretic attributions	Backpropagated gradients to final conv layer
Visualization	Feature contribution bar chart	Pixel-aligned visual heatmap overlay
Primary Objective	Transparent triage justification	Vascular lesion verification & confirmation
Clinical Impact	Increases screening compliance	Prevents false-positive laser referrals

Technical Specifications & Future Backend Roadmap

Implementation architecture, multi-language engine, and database scaling

Multi-Language & Adaptive Theme Specifications

? Trilingual Localization Engine: Built using a modular dictionary architecture in `src/lib/translations.js`. Supports English (en), Marathi (mr - ?????), and Hindi (hi - ?????). Switching languages instantly re-renders all labels, form placeholders, risk outputs, diagnosis descriptions, and portal headers across the entire platform.

? Aptos Light Typography: Applied universally across all screens. Avoids harsh gradient text fills, prioritizing high-contrast solid colors (black, deep purple, and crisp white) for medical clarity.

? Adaptive Theme Architecture:
- Light Mode: Light purple background (#f1e8fc) paired with frosted white cards and thin purple borders.
- Dark Mode: Preserves deep black-purple tones (#0b0416) with subtle purple glassmorphic borders.
- Corner Accents: Ambient curved light shapes in page corners for a modern, approachable aesthetic.

Future Backend & AI Model Training Roadmap

When transitioning from client-side LocalStorage to a production backend infrastructure, the recommended architecture is:

1. Database Selection:
? PostgreSQL (via Supabase or Managed RDS): Highly recommended. Provides ACID compliance, relational foreign keys between patients, appointments, and imaging records, and built-in Row-Level Security (RLS) for HIPAA compliance.
? MongoDB: Alternative for document-oriented storage if imaging metadata and flexible symptom trees expand rapidly.

2. Server Framework: Python FastAPI or Node.js Express/NestJS:
? FastAPI provides native asynchronous endpoints, automatic OpenAPI documentation, and seamless PyTorch/TensorFlow integration.

3. AI Model Training on Custom Indian Clinic Datasets:
? Datasets: Combine open-source benchmarks (EyePACS, Messidor-2, APTOS 2019) with locally annotated rural clinic scans.
? Architecture: EfficientNet-B4 or Vision Transformer (ViT) pre-trained on ImageNet, fine-tuned with focal loss for DR grading.
? Explainability: Hook PyTorch backward hooks to extract activation maps at the final convolutional layer for Grad-CAM.
? Export: Quantize to ONNX / TensorRT for edge inference on low-power hospital laptops.

Clinical Quality Assurance & Governance Checklist

- Patient Consent & Privacy: All patient data gathered during rural visits requires informed verbal/written consent.
- Image Quality Assurance: Nurse portal validates fundus illumination, focus, and field of view before model inference.
- Human-in-the-Loop Supremacy: AI outputs are purely assistive. Final clinical diagnosis remains the sole responsibility of the ophthalmologist
- Audit Logging: Every report modification, status update, and dispatch action generates an immutable audit timestamp.
- Offline Continuity: Field survey modules retain local caching to ensure uninterrupted operation during rural connectivity drops.

Approved for Clinical Demonstration & Architectural Review